Chemometrics Web App: Synthetic Sampling (CWA: SS)

Comprehensive Step-by-Step User Guide & Tutorial Manual

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1. Introduction & overview

The Chemometrics Web App: Synthetic Sampling (CWA: SS) is an open-source, browser-accessible graphical software designed to overcome the persistent challenge of class imbalance in chemometric and analytical chemistry datasets. In field chemistry, food authentication, and quality control, class imbalance occurs naturally. The phenomena or samples of primary scientific interest, such as adulterated products, rare toxic contaminants, or early-stage material degradation, are inherently far scarcer than routine, compliant majority samples. When multivariate predictive discriminant models are trained on such unbalanced data, they frequently exhibit a classic statistical trap: a deceptively high overall accuracy masking a total collapse in sensitivity toward the minority class. In practice, this means the model fails precisely where its detection capabilities are most critical.

To resolve this, CWA: SS integrates the flexible user interface of R/Shiny with the robust algorithmic backend of Python's `imbalanced-learn` library (https://imbalanced-learn.org/stable/), established via the reticulate package (https://cran.r-project.org/package=reticulate). This software unifies 16 advanced resampling algorithms into a single, installation-free web environment. Crucially, CWA: SS introduces a rigorous diagnostic and quality control module. Rather than trusting synthetic data blindly, the app provides a comprehensive suite of multivariate statistical tests and automated visual evaluations. The tools implemented in this interface ensure that the generated synthetic samples accurately preserve the underlying physical, chemical, and mathematical covariance of the original systems, preventing the introduction of unphysical artifacts before data is fed into downstream machine learning models.

CWA: SS is the third module in the Chemometrics Web App (CWA) series, following CWA: Data Handling & Preprocessing (https://doi.org/10.1016/j.chemolab.2022.104696) and CWA: Exploratory Analysis & Dimensionality Reduction (https://doi.org/10.1016/j.chemolab.2023.104810).

2. Software access & execution modes

The Chemometrics Web App: Synthetic Sampling (CWA: SS) is distributed across three complementary execution modes to accommodate different analytical workflows, hardware configurations, and user technical backgrounds. Users can access the platform via: (1) an installation-free cloud web application; (2) a pre-compiled standalone Windows desktop executable (.exe); or (3) local source-code execution within RStudio (runApp).

2.1 Mode 1 - Cloud-hosted web application (shinyapps.io)

• Access URL: https://ltap.shinyapps.io/Synthetic_Sampling/

• Prerequisites & Setup: 100% installation-free. Compatible with any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera) across Windows, macOS, Linux, ChromeOS, iPadOS, iOS, and Android.

• Step-by-Step Usage:

1. Open the URL in any standard web browser.

2. The interface loads immediately into Tab 1 (Data Import and Pre-processing).
3. Upload custom data (.csv, .xlsx, .txt) or click 'Use Demo Dataset' (QSAR or Prestige) to start exploring.
4. Proceed sequentially through Tabs 2 to 4 to balance data and download final augmented matrices.

• Recommended Use Cases: Interactive demonstrations, academic classroom teaching, rapid preliminary assessments in the field via mobile devices, and standard multivariate datasets.

• Infrastructure Notes: The cloud version is hosted on shinyapps.io servers with an 8 GB RAM allocation and monthly active connection quotas. For high-dimensional spectral matrices (> 1,000 variables) or computationally intensive resampling (SPIDER, SMOTE-IPF, t-SNE), the standalone desktop version or local RStudio execution is strongly recommended.

Access Mode	Link
Cloud Web Application	https://ltap.shinyapps.io/Synthetic_Sampling/
Standalone Desktop (.exe, Windows)	https://www.ltapuerj.com.br/cwa or directly https://drive.google.com/drive/folders/1l8dB4BGKVjqPMrvA5ZCLvafRNtTOCFew?usp=drive_link
Public Mirror	https://sites.google.com/view/ltap-uerj/cwa
GitHub Source Code Repository	https://github.com/LTAP-UERJ
Community Feedback Form	https://forms.gle/D7mkkRBqQsvX9ZhW6

2.2 Mode 2 — Standalone Windows desktop executable (.exe)

• Download Links:

- Official LTAP Laboratory Website: https://www.ltapuerj.com.br/cwa
- Direct Google Drive Link: https://drive.google.com/drive/folders/1l8dB4BGKVjqPMrvA5ZCLvafRNtTOCFew?usp=sharing
- GitHub Repository: https://github.com/LTAP-UERJ/Chemometrics-Web-App-Synthetic_Sampling

• Prerequisites & System Requirements: Windows 10 or Windows 11 (64-bit). Minimum 4 GB RAM (8 GB or more recommended). No prior installation of R, RStudio, or Python is required on the host system.

• Step-by-Step Installation & Execution Guide:

1. Download the setup installer (Synthetic_Sampling_App_Setup.exe) from the official links above.
2. Run the setup installer and follow the on-screen wizard (standard user-level installation without administrative rights).

3. The installer creates a desktop shortcut and a Start Menu entry named 'Synthetic Sampling App'.
4. Double-click the desktop shortcut. The software automatically launches its embedded, self-contained R-Portable and Python environments with all 41 required packages pre-installed (including shinyBS, shinydashboard, mdatools, vegan, Hotelling, reticulate, scikit-learn, and imbalanced-learn).
5. The application will open automatically in your default web browser at localhost (http://127.0.0.1:XXXX).

• Recommended Use Cases: Processing large continuous spectroscopic matrices (NIR, FTIR, Raman), intensive resampling algorithms, air-gapped or offline laboratory workstations, and high-performance chemometric research without cloud memory limits.

2.3 Mode 3 - Local source code execution in RStudio (runApp / GitHub)

• GitHub Repository: https://github.com/LTAP-UERJ/Chemometrics-Web-App-Synthetic_Sampling
• Prerequisites: R (version >= 4.2.0) and RStudio installed on Windows, macOS, or Linux.
• Step-by-Step Setup & Execution Guide:

1. Download or clone the repository:

   git clone https://github.com/LTAP-UERJ/Chemometrics-Web-App-Synthetic_Sampling.git

   (or download and extract the repository ZIP file).

2. Automated Dependency Installation:

   Open RStudio, set the working directory to the project folder, and run:

   source('install_dependencies.R')

   (or execute install_cwa_synthetic_sampling_packages()).

   The script automatically checks, installs, and verifies all 41 required CRAN packages (shinyBS, shinydashboard, mdatools, vegan, Hotelling, smotefamily, UBL, scutr, reticulate) and configures the Python virtual environment with scikit-learn (module sklearn) and imbalanced-learn (module imblearn).

3. Launch the Application:

   In RStudio, open app.R (or Synthetic_Sampling_App.R) and click the 'Run App' button in the editor toolbar, or type in the RStudio console:

   shiny::runApp()

   The application interface will immediately launch in the RStudio Viewer pane or your default web browser.

• Recommended Use Cases: Cross-platform execution on macOS and Linux, custom algorithmic development and modification, integration into automated chemometrics scripts, and full research reproducibility.

3. System requirements & architecture

Cloud Version: Compatible with any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) on Windows, macOS, Linux, Android, or iOS. No installation required.

Standalone Desktop Version: Windows 10/11 (64-bit). Minimum 4 GB RAM; 8 GB or more recommended for spectral matrices with > 1,000 variables or > 500 samples. A bundled Python 3.x environment with all required packages is pre-installed in the portable runtime.

Local Execution from Source (GitHub / Cross-Platform): Users operating on Windows, macOS, or Linux who prefer running from source can clone our GitHub repository (https://github.com/LTAP-UERJ/Chemometrics-Web-App-Synthetic_Sampling). To automate the configuration of all R and Python packages, a dedicated script (install_dependencies.R) is provided in the repository root. Simply open R/RStudio and run source('install_dependencies.R') or execute install_cwa_synthetic_sampling_packages(). This function automatically checks, installs, and validates all required CRAN packages (including shinyBS, shinydashboard, mdatools, vegan, Hotelling, reticulate) and configures the Python virtual environment with scikit-learn and imbalanced-learn.

Computational Architecture: CWA: SS uses a hybrid dual-language engine:

Component	Technology	Role
User Interface	R/Shiny + shinydashboard	GUI rendering, interactivity, visualization
Visualization	plotly, ggplot2, DT	Interactive charts and responsive tables
Native Oversampling	smotefamily (R)	SMOTE, Borderline-SMOTE (no Python required)
Python-based methods	imbalanced-learn, scikit-learn	ADASYN, SVM-SMOTE, SMOTE-NC, Tomek, NearMiss, ENN, OSS, RD, SBC, SMOTE-TL, SMOTE-ENN, SMOTE-IPF, SPIDER
Bridge	reticulate	Bidirectional R ↔ Python object transfer
Multivariate Analysis	mdatools, vegan, Hotelling, Rtsne, philentropy	PCA, ROBPCA, t-SNE, PERMANOVA, Hotelling T^2, KS test, JS Divergence, GPA

4. Software navigation & core modules

The CWA: SS interface is organized sequentially into five main tabs. The recommended analytical workflow is to proceed from left to right:

Tab 1: Data Import & Pre-processing → Tab 2: Sampling Methods → Tab 3: Diagnostic & QC → Tab 4: Results & Export → Tab 5: Authors & References

4.1 Tab 1: Data Import and Pre-processing

This tab handles all data ingestion, inspection, and optional pre-processing before resampling is applied.

Step 1 — File Upload

Click the Upload Data File button and select a dataset in one of the accepted formats: .csv, .txt, .xls, or .xlsx. The application automatically detects the separator (comma, semicolon, or tab) and displays a preview of the first rows and columns. Alternatively, click "Use Demo Dataset" to load one of the two demonstration files pre saved in the graphical user interface.

Dataset input requirements:

Requirement	Specification
Row orientation	Each row = one observation (sample)
Column orientation	Each column = one variable (descriptor or wavelength)
Class column	One dedicated column containing the class label (text or integer)
Missing values	Minimal (NaN handling recommended before upload)
Variable types	Numeric variables for most methods; SMOTE-NC accepts mixed types

Step 2 — Target Variable Selection

Select the column containing class labels using the target variable dropdown. The application displays a class frequency bar chart, clearly indicating which classes are in the minority and which are in the majority.

Step 3 — Data Cleaning (Optional)

CWA: SS provides light pre-processing utilities:

- **Remove Zero-Variance Columns:** Eliminates descriptor columns with no discriminating information (standard deviation = 0).
- **Remove Duplicate Rows:** Identifies and optionally removes exact duplicate observations.
- **Remove Missing Values:** Drops rows or columns with missing values above a defined threshold.

Step 4 — Preprocessing (Optional)

Apply standard chemometric scaling to remove unit and magnitude disparities between descriptors:

Method	Formula	When to Use
Mean Centering	$x' = x - \bar{x}$	When variables share common units; removes mean offset
Autoscaling	$x' = (x - \bar{x}) / s$	When variables have different units or dynamic ranges; essential for k-NN-based methods
Min-Max Normalization	$x' = (x - x_{min}) / (x_{max} - x_{min})$	When bounded [0, 1] range is required

Log Transform	*x′ = log(x + 1)*	For right-skewed count or intensity data
Yeo-Johnson Transform	Power transform optimizing normality	For general skewness correction including zero/negative values
Box-Cox Transform	Power transform for strictly positive data	Specialized normality correction for positive measurements

Important: Preprocessing must be applied *before* running any resampling algorithm. All 16 algorithms in Tab 2 operate on the preprocessed dataset.

4.2 Tab 2: Resampling Methods

Navigate to the desired algorithm using the sidebar. Each method panel is self-contained and includes:

1. An educational description of the algorithm, its optimal use case, and a chemometric application example.
2. Input controls for all method-specific parameters.
3. A run method button to execute the calculation.
4. A confirm button that incorporates the synthetic or balanced dataset for use in Tabs 3 and 4.
5. An interactive result bar chart comparing class counts before and after resampling.

Important: Only one resampling method can be active at a time. Running a second method automatically resets the previously confirmed result. Sequential application of distinct methods (e.g., SMOTE followed by SBC) is supported.

4.3 Tab 3: Diagnostic and quality control module

The diagnostic module evaluates the fidelity of synthetic samples through a comprehensive suite of visual and statistical tools. This tab is accessible only after a method has been confirmed in Tab 2.

Two execution modes are available:

- "Plot All Charts & Run All Tests": Executes all 11 diagnostic tools sequentially in a single action, with a progress bar indicating each step.
- Individual diagnostic buttons: Each diagnostic section has its own run button for targeted single-output execution.

Details on interpretation criteria for each diagnostic are provided in Section 6 of this guide.

4.4 Tab 4: Results and export

- Combined Dataset Preview: Displays the full balanced dataset (Original + Synthetic/Removed rows), with a Type column labeling each row.
- Method Summary Table: Reports the final class distribution, number of synthetic samples added, and number of samples removed per class.

- Download buttons:
- Download as excel (.xlsx): Produces a structured spreadsheet with sample labels, class assignments, and all numeric variables.
- Download as CSV (.csv): Produces a flat-file format compatible with R, Python, MATLAB, and other external modeling tools.
- Analytical report (PDF): Generates a compiled PDF report including method summary, diagnostic plots, and statistical test results.

5. Algorithm parameter reference guide

The following tables describe every configurable parameter exposed by CWA: SS for each of the 16 algorithms.

5.1 Oversampling Methods

SMOTE — Synthetic Minority Over-sampling Technique

Parameter	Type	Default	Description
k_neighbors	Integer	5	Number of nearest neighbors used to define the interpolation neighborhood. Lower values (3–5) reduce noise; higher values (7–10) produce greater diversity. Must be < minority class size.
dup_size	Integer	1	Number of synthetic samples to generate per original minority sample. dup_size = 1 doubles the minority class; dup_size = 2 triples it.
Target Class	Categorical	—	The specific class to oversample. When "All Classes" is selected, SMOTE balances all minority classes toward the majority simultaneously.

The SMOTE algorithm interpolates new samples along the line segment connecting a minority sample to one of its k randomly selected nearest neighbors:

$$x\_new = x\_i + \lambda \cdot (x\_knn - x\_i), \text{ where } \lambda \sim \text{Uniform}(0, 1)$$

SMOTE-NC — SMOTE for Nominal and Continuous Features

Parameter	Type	Default	Description
k_neighbors	Integer	5	Neighbors considered for mixed-type interpolation.
Categorical Columns	Multi-select	—	Columns to treat as categorical. Categorical features are handled via mode imputation during interpolation.
sampling_strategy	Float or "auto"	auto	Target ratio of minority to majority samples. "auto" balances all classes to the majority count.

Borderline-SMOTE

Parameter	Type	Default	Description
k_neighbors	Integer	5	Neighbors used to determine whether a minority sample lies on the borderline (danger zone).
m_neighbors	Integer	10	Neighbors examined to classify each minority sample as safe, danger, or noise.
kind	Select	borderline-1	borderline-1: only border-zone samples are synthesized. borderline-2: includes majority-class samples in the interpolation axis.
sampling_strategy	Float or "auto"	auto	Target sampling ratio.

Selection criterion: A minority sample is classified as "danger" (eligible for synthesis) if more than half but fewer than all of its *m* neighbors belong to the majority class.

SVM-SMOTE — Support Vector Machine SMOTE

Parameter	Type	Default	Description
k_neighbors	Integer	5	Neighbors used for synthesis around support vectors.
m_neighbors	Integer	10	Neighbors for borderline detection.
sampling_strategy	Float or "auto"	auto	Target ratio.

Note: An internal SVM with RBF kernel is trained to identify support vectors (minority samples closest to the decision boundary). Synthesis is concentrated around these boundary-proximal samples.

ADASYN — Adaptive Synthetic Sampling

Parameter	Type	Default	Description
k_neighbors	Integer	5	Neighbors used to compute local density ratios. More neighbors = smoother density estimates.
sampling_strategy	Float or "auto"	auto	Overall target ratio.

ADASYN computes a density ratio r̂_i for each minority sample x_i: the proportion of majority-class samples among its k-nearest neighbors. The number of synthetic samples generated at x_i is proportional to r̂_i, generating more samples in harder-to-learn, minority-sparse regions.

Random Upsampling (RU)

Parameter	Type	Default	Description
Target Class	Categorical	—	Class to replicate.
sampling_strategy	Float or "auto"	auto	Target ratio of minority to majority.

Note: RU does not generate new data points. It replicates existing minority observations. Recommended only when sample sizes are very small and k-NN-based methods are numerically unstable.

5.2 Undersampling Methods

Tomek Links

Parameter	Type	Default	Description
sampling_strategy	String	"not minority"	Specifies which class(es) to clean. Default removes ambiguous majority samples near the boundary.

A Tomek Link exists between samples x_i and x_j of different classes if they are each other's nearest neighbor: NN(x_i) = x_j and NN(x_j) = x_i. Removing the majority member of such pairs cleans the decision boundary.

NearMiss Undersampling

Parameter	Type	Default	Description
version	Integer	1	v1: keeps majority samples closest to the *nearest* minority samples. v2: keeps majority samples closest to the *farthest* minority samples. v3: for each minority sample, retains the *m* majority neighbors closest to it.
n_neighbors	Integer	3	Number of neighbors considered in the selection criterion.
sampling_strategy	Float or "auto"	auto	Target final ratio.

Edited Nearest Neighbors (ENN)

Parameter	Type	Default	Description
n_neighbors	Integer	3	Number of neighbors for the cleaning rule. A majority sample is removed if the majority of its *k* neighbors belong to a different class.
kind_sel	Select	all	all: removes a sample if *all* neighbors disagree with its class. mode: removes if the *majority* of neighbors disagree.

One-Sided Selection (OSS)

OSS applies two consecutive steps: (1) Tomek Links removal, then (2) Condensed Nearest Neighbor (CNN) undersampling. No user-adjustable parameters beyond sampling_strategy.

Random Downsampling (RD)

Parameter	Type	Default	Description
Target Class	Categorical	—	Class to undersample.
sampling_strategy	Float or "auto"	auto	Target post-removal ratio.
random_state	Integer	42	Reproducibility seed.

Cluster-Based Undersampling (SBC)

Parameter	Type	Default	Description
n_clusters	Integer	8	Number of *k*-means clusters to partition the majority class. Each cluster is replaced by its centroid.
Target Classes	Multi-select	—	Majority class(es) to undersample.

SBC preserves the geometric diversity of the majority class by replacing dense clusters of samples with their centroids, maintaining the overall distribution shape with fewer data points.

5.3 Hybrid Methods

SMOTE-TL (SMOTE + Tomek Links)

First applies SMOTE oversampling, then applies Tomek Links to remove ambiguous synthetic samples near class boundaries. Parameters: SMOTE (k_neighbors, dup_size) + Tomek Links (sampling_strategy).

SMOTE-ENN (SMOTE + Edited Nearest Neighbors)

First applies SMOTE oversampling, then applies ENN to remove noisy or misclassified synthetic samples. More aggressive cleaning than SMOTE-TL. Parameters combine SMOTE and ENN controls.

SMOTE-IPF (SMOTE + Iterative Partitioning Filter)

Parameter	Type	Default	Description
k_neighbors	Integer	5	SMOTE neighborhood.
n_estimators	Integer	10	Number of estimators in the IPF ensemble. Each trains on a data partition to identify noisy instances.
n_iters	Integer	3	Number of filtering iterations. Higher values provide more aggressive noise removal.

SPIDER (Selective Preprocessing of Imbalanced Data)

Parameter	Type	Default	Description
k_neighbors	Integer	5	Neighbors used to classify each sample as safe, borderline, or noisy.
delta	Float	0.5	Threshold for the safety decision boundary. Higher values increase the borderline region.

SPIDER classifies minority samples into three categories: *safe* (reinforced), *borderline* (oversampled), and *noisy* (removed along with nearby majority samples). It is particularly effective for datasets where noise and class overlap co-occur.

6. Diagnostic Interpretation Guide: Acceptance and Rejection Criteria

The diagnostic module provides the following tools. For each, acceptance criteria (indicating the resampling preserved data integrity) and rejection criteria (indicating artifacts or distribution shifts) are specified.

6.1 Class Frequency Bar Chart

Purpose: Confirms that the resampling achieved the desired class balance.

Criterion	Acceptable Result	Problematic Result
Class counts post-resampling	All classes have comparable sample sizes or achieve target ratio	One class remains severely under-represented
Synthetic sample proportion	< 200% of original minority size	> 300% suggests excessive oversampling

6.2 Variable Mean Profile Plot

Purpose: Verifies that the mean descriptor profile of synthetic samples matches that of original samples, confirming no systematic signal distortion.

Criterion	Acceptable Result	Problematic Result
Mean profile overlap	Synthetic mean profile overlays closely with original (< 5% relative difference per variable)	Systematic offset, peak shifts, or flipped features
Baseline consistency	Flat residual between mean profiles	Non-zero structured baseline difference

6.3 Individual Sample Profile Overlay

Purpose: For spectral data (NIR, Raman, UV-Vis), verifies signal continuity of synthetic spectra.

Criterion	Acceptable Result	Problematic Result
Signal continuity	Smooth, physically interpretable absorption/emission curves	High-frequency noise, discontinuities, or unphysical spikes
Feature position	Absorption peaks at same wavelengths as original spectra	Shifted peak positions

6.4 PCA and Robust PCA Score Plots (2D and 3D)

Purpose: Visualizes the distribution of synthetic samples within the original chemical space.

Criterion	Acceptable Result	Problematic Result
Spatial overlap	Synthetic points embedded within or adjacent to original cluster	Synthetic points form a separate cluster or extend beyond original chemical space
Hotelling T^2 ellipse	≥ 95% of synthetic points inside the 95% confidence ellipse	More than 5% of synthetic points outside the ellipse
Within-class homogeneity	Synthetic samples visually indistinguishable from original within their class	Synthetic samples form a distinct sub-cluster within their class

6.5 PCA Residual Diagnostics (Q vs. T² Plot)

Purpose: Detects outlier synthetic samples using PCA leverage and orthogonal distance.

Criterion	Acceptable Result	Problematic Result
Leverage (T^2)	Synthetic samples have T^2 values comparable to original samples	Synthetic samples show anomalously high T^2 (extreme leverage)
Orthogonal distance (Q)	Synthetic samples fall within the same Q distribution as original samples	Elevated Q values indicate samples not captured by the PCA model

6.6 PERMANOVA (Permutation Multivariate Analysis of Variance)

Purpose: Tests the null hypothesis that the multivariate distributions of original and synthetic samples from the same class are statistically identical.

Implementation: adonis2 function (vegan package, R). Permutations: 499 for standard datasets; 199 for large datasets ($N > 500$ samples or > 5 classes) to preserve computational feasibility.

Criterion	Acceptable	Problematic
p-value	$p > 0.05$ — fail to reject H_0; distributions are statistically equivalent	p_{eq} *0.05* — distributions differ significantly
F-statistic	Values near 1 (no between-group separation)	High *F* values (strong between-group separation)

6.7 Hotelling's T^2 Test (Multivariate Mean Equality Test)

Purpose: Tests whether the multivariate centroid (mean vector) of the synthetic sample group equals that of the original sample group within the same class.

Implementation: Manual Hotelling T^2 calculation with regularization ($\delta = 10^{-5}$) and pseudo-inverse fallback (MASS::ginv) for singular or near-singular pooled covariance matrices arising from high-dimensional or collinear data:

$$T^2 = (n_1 \cdot n_2 / (n_1 + n_2)) \cdot (\bar{x}_1 - \bar{x}_2)^T \cdot S\_pooled^{-1} \cdot (\bar{x}_1 - \bar{x}_2)$$

Criterion	Acceptable	Problematic
p-value	$p > 0.05$ — mean vectors are not significantly different	p_{eq} *0.05* — synthetic centroid has shifted from original

6.8 Kolmogorov-Smirnov (KS) Test

Purpose: Univariate distribution test applied to each descriptor individually. Tests whether the empirical cumulative distribution functions of original and synthetic samples are statistically equivalent.

Criterion	Acceptable	Problematic
p-value per variable	$p > 0.05$ for the majority of variables	p_{eq} *0.05* for more than 20% of variables
Result summary	Most variables labeled "No Significant Difference"	Multiple variables flagged as significantly different

Note: For high-dimensional datasets (e.g., full-spectrum NIR with 256 wavelengths), a small number of rejected variables (< 10%) is acceptable by chance. A practical threshold for concern is > 20% of variables rejected.

6.9 Jensen-Shannon (JS) Divergence

Purpose: Quantifies the probabilistic similarity between the joint distributions of original and synthetic samples. JS Divergence ranges from 0 (identical distributions) to 1 (completely divergent distributions). It is a symmetric, bounded variant of the Kullback-Leibler divergence:

$$JS(P \| Q) = \tfrac{1}{2} \cdot KL(P \| M) + \tfrac{1}{2} \cdot KL(Q \| M), \text{ where } M = (P + Q) / 2$$

Criterion	Excellent	Acceptable	Problematic
JS Divergence	< 0.05	0.05 – 0.15	> 0.15

6.10 Generalized Procrustes Analysis (GPA)

Purpose: Measures geometric consensus between the original and augmented dataset matrices in reduced PCA space. GPA superimposes two configurations after optimal rotation, scaling, and translation, and computes a residual sum-of-squares.

Criterion	Acceptable	Problematic
Procrustes residual	Low residual (near 0); configurations overlap substantially	High residual; marked structural mismatch
GPA scatter plot	Short, uniform arrows between matched pairs	Long or systematically directed arrows indicate structural distortion

6.11 Variance Equality Test (Levene's Test)

Purpose: Tests whether synthetic samples have similar within-class variance structure compared to original samples for each descriptor.

Criterion	Acceptable	Problematic
p-value per variable	$p > 0.05$ — equal variances	p_{eq} 0.05 — inflated or deflated variance in synthetic set

7. Step-by-Step guided tutorials with demo files

To help users familiarize themselves with the workflow, CWA: SS includes two built-in demonstration datasets. These files allow you to explore the software's capabilities immediately without uploading external data. Both datasets can be accessed directly via Tab 1 using the "Use Demo Dataset" dropdown, or downloaded for offline use from the official repository (https://github.com/LTAP-UERJ/Chemometrics-Web-App-Synthetic_Sampling).

Tutorial 1: QSAR Bioconcentration Dataset (Discrete molecular descriptors)

Dataset: QSAR Dataset (bundled) freely available at https://archive.ics.uci.edu/dataset/254/qsar+biodegradation.

Background: This dataset contains 779 chemical compounds described by 9 physicochemical molecular descriptors (e.g., molecular weight, LogP, molar refractivity, topological polar surface area). Compounds are classified into three bioconcentration classes based on their experimentally measured fish bioconcentration factor (BCF): Class 1 (*High BCF*, $n = 460$), Class 2 (*Moderate BCF*, $n = 64$), and Class 3 (*Low BCF*, $n = 255$). The severe imbalance in Class 2 makes this a challenging classification problem.

Objective: Apply a sequential hybrid strategy (SMOTE + SBC) to balance all three classes while preserving the chemical descriptor space.

Step 1 — Load the Demo Dataset

1. Navigate to Tab 1 "Data Import & Pre-processing".
2. Click "Use Demo Dataset" and select QSAR Dataset.
3. The application loads the 779-sample matrix and displays the class frequency bar chart.
4. Confirm the imbalance: Class 1 (n = 460), Class 2 (n = 64), Class 3 (n = 255).
5. Set the Target Variable dropdown to "Class".

Step 2 — Apply Autoscaling

6. In the Preprocessing section, select "Autoscaling".
7. Click "Apply Preprocessing".
8. Verify that the data summary table now shows standardized values (means ≈ 0, standard deviations ≈ 1 for all variables).

Step 3 — Oversample Class 2 with SMOTE

9. Navigate to Tab 2 "Sampling Methods".
10. Select "SMOTE" from the sidebar.
11. Set Target Class to "Class 2".
12. Set k_neighbors = 5 and dup_size = 2 (will generate 128 synthetic samples, tripling Class 2 from 64 to 192 observations).
13. Click "Run SMOTE".
14. Review the result bar chart confirming the new Class 2 count.
15. Click "Confirm SMOTE" to incorporate the synthetic samples.

Step 4 — Undersample Class 1 with Cluster-Based Undersampling (SBC)

16. Without leaving Tab 2, select "Cluster-Based Undersampling (SBC)" from the sidebar.
17. Set Target Classes to "Class 1".
18. Set n_clusters = 8.
19. Click "Run SBC" then "Confirm SBC".

Expected final distribution: Class 1 ≈ 200-250, Class 2 ≈ 192, Class 3 = 255 (unchanged).

Step 5 — Gatekeeper Diagnostic Verification

20. Navigate to Tab 3 "Diagnostic and Quality Control".
21. Click "Plot All Charts & Run All Tests". A progress bar indicates sequential calculation across all 11 diagnostic outputs.
22. Inspect the following expected acceptable results:

Diagnostic	Expected Result	Interpretation
Variable Mean Profile	Synthetic profiles overlay original profiles	No descriptor distortion
PCA Score Plot (PC1 vs PC2)	Synthetic Class 2 points cluster within original Class 2 region	Preserved chemical space
PERMANOVA (Class 2)	$p > 0.05$	Synthetic distribution not significantly different
Hotelling T^2 (Class 2)	$p > 0.05$	Centroid preserved
KS Test	Most descriptors $p > 0.05$	Univariate distributions maintained
JS Divergence	< 0.10	High distribution fidelity

Step 6 — Export the Balanced Dataset

23. Navigate to Tab 4 "Results and Export".
24. Review the combined dataset table confirming rows are labeled "Original", "Synthetic", or "Removed" in the Type column.
25. Click "Download as Excel (.xlsx)" to save the balanced dataset for downstream classification modeling.

Tutorial 2: Prestige Dataset (Mixed tabular data)

Dataset: Prestige Dataset, freely available at https://www.stat.auckland.ac.nz/~wild/data/Rdatasets/.

Background: This dataset contains occupational prestige scores and socioeconomic indicators (income, education, women participation percentage) for 102 occupational categories in Canada, classified into three groups: blue-collar (*bc*, $n = 44$), professional (*prof*, $n = 31$), and white-collar (*wc*, $n = 23$). White-collar occupations represent the minority class in this low-sample imbalanced scenario.

Objective: Demonstrate the software's functionality on a non-spectral tabular dataset, applying SMOTE to the minority class and validating quality with the full diagnostic suite.

Step 1 — Load the Prestige Demo Dataset

1. Navigate to Tab 1 "Data Import & Pre-processing".
2. Click "Use Demo Dataset" and select Prestige Dataset.
3. Confirm the loaded matrix: 102 rows × 5 columns + 1 class column.
4. Set the Target Variable to "type".
5. Note the imbalance: *bc* ($n=44$) vs. *prof* ($n=31$) vs. *wc* ($n=23$).

Step 2 — Apply Mean Centering

6. Select "Auto Scaling" in the Preprocessing panel.
7. Click "Apply Preprocessing".

Step 3 — Oversample White-Collar (wc) with SMOTE

8. Navigate to Tab 2 "Sampling Methods" → "SMOTE".
9. Set Target Class = "wc".
10. Set k_neighbors = 4 (adjusted for small class; must be < 23) and dup_size = 1.
11. Click "Run SMOTE" → "Confirm SMOTE".

Expected result: White-collar class grows from 23 to 46 observations, comparable to blue-collar.

Step 4 — Diagnostic Verification for Small-Sample Context

12. Navigate to Tab 3 "Diagnostic and Quality Control".
13. Click "Plot All Charts & Run All Tests".
14. For this small-sample dataset, pay attention to:

- PCA Score Plot: With only 4 descriptors, PC1+PC2 should capture > 85% of variance. Confirm synthetic *wc* points are embedded within the original *wc* cluster.
- Hotelling's T^2 Test: With n = 23 original + n = 23 synthetic, a non-significant result ($p > 0.05$) confirms centroid preservation.
- KS Test: With 4 variables, all should remain non-significant if the synthesis is well-constrained.

Step 5 — Export

15. Navigate to Tab 4 "Results and Export".
16. Download the balanced dataset in .csv format for use in external classification workflows (e.g., LDA, PLS-DA, SVM, or Random Forest).

8. Chemometric best practices for synthetic sampling

The following guidelines encode community consensus adapted specifically for analytical chemistry and chemometrics applications.

1. **Mandatory test set isolation.** Synthetic samples must never be included in the validation or test set. The test set must consist exclusively of real, experimentally acquired samples. All resampling must be confined to the training set within any cross-validation pipeline.
2. **Preprocessing before resampling.** Apply autoscaling or mean-centering before running any k-NN-based algorithm. Unscaled variables with vastly different dynamic ranges will produce biased nearest-neighbor calculations and geometrically inconsistent synthetic samples.
3. **Moderate oversampling.** Avoid inflating a minority class beyond 3 times its original size (dup_size > 2). Excessive oversampling with small neighborhoods (k = 3) can produce a degenerate synthetic cloud with inflated density far from true data.

4. **Prefer hybrid methods for overlapping classes.** For datasets with inherent class overlap, a pure oversampling approach amplifies noise at the boundary. Use hybrid methods (SMOTE-TL, SMOTE-ENN, SPIDER) to simultaneously synthesize and clean.
5. **Require all diagnostic tests to pass.** A resampling result should not be accepted for downstream modeling if the PERMANOVA p-value is significant ($p < 0.05$) or if JS Divergence exceeds 0.15. In these cases, reduce dup_size, increase k_neighbors, or switch to a different algorithm.
6. **Document all parameter choices.** For reproducibility and publication transparency, record the exact algorithm, parameter values, and random seed used. CWA: SS automatically includes this information in the exported PDF Analytical Report.
7. **Sequential oversampling + undersampling is valid.** It is scientifically appropriate to apply an oversampling method to the minority class followed by an undersampling method to the majority class in sequence. CWA: SS fully supports sequential application of distinct methods.

9. Troubleshooting & Frequently Asked Questions (FAQ)

9.1 Data Import

Q: My file uploads but no data appears in the preview.

A: Verify that the file uses a consistent column separator (comma, semicolon, or tab). If using .xlsx, ensure no merged cells are present in the header row. Remove any trailing empty columns or rows before uploading.

Q: The target variable dropdown shows no options.

A: Check that your class column contains text labels or integer codes. The column must have at least 2 unique values. If all columns appear numeric, verify that class labels were not encoded as numeric factors.

Q: I receive an error "not enough samples to apply SMOTE".

A: The selected minority class has fewer samples than k_neighbors + 1. Reduce k_neighbors to at least 1 less than the minority class size (e.g., k_neighbors = 2 for a class with 3 samples), or switch to Random Upsampling (RU) which has no minimum sample requirement.

9.2 Algorithm execution

Q: ADASYN, SVM-SMOTE, or SMOTE-NC returns "Python library not found".

A: These methods require the Python imbalanced-learn package. In the cloud version, all Python dependencies are pre-installed. In the standalone desktop version (.exe), Python libraries and R-Portable dependencies (including shinyBS) are pre-packaged. For local source execution, run source('install_dependencies.R') in RStudio, which automatically configures reticulate and installs the required Python modules (imbalanced-learn and scikit-learn). If using an existing virtualenv, verify that reticulate::py_module_available('imblearn') returns TRUE.

Q: SMOTE runs but generates 0 synthetic samples.

A: This occurs when dup_size = 0 or when all minority samples are identical (zero variance in all variables). Verify that dup_size ≥ 1 and that the minority class contains at least 2 unique observations.

Q: The application shows a loading spinner for more than 2 minutes without completing.

A: This may occur in the cloud version with large datasets (> 500 samples and > 100 variables) when running computationally expensive methods (SPIDER, SMOTE-IPF, t-SNE). Switch to the standalone desktop version for large-scale analyses, which has no RAM or timeout restrictions.

Q: Running all diagnostics simultaneously causes the interface to freeze in the cloud version.

A: Use the individual diagnostic buttons to run calculations one at a time instead of the batch button, or use the desktop version.

9.3 Diagnostic Interpretation

Q: The PERMANOVA p-value is < 0.05 for my dataset. What should I do?

A: A significant PERMANOVA result indicates the resampling introduced a statistically detectable distributional shift. Recommended actions: (1) increase k_neighbors to produce more diverse synthetic samples; (2) reduce dup_size to avoid oversaturation; (3) switch algorithms, e.g., Borderline-SMOTE instead of standard SMOTE for high-overlap datasets.

Q: The PCA score plot shows synthetic points forming a separate sub-cluster. Is this acceptable?

A: No. A distinct synthetic sub-cluster indicates the algorithm is extrapolating beyond the original data manifold, a common artifact of SMOTE with very small k_neighbors in sparse data. Increase k_neighbors or reduce dup_size.

Q: The Hotelling T^2 test reports "Error: Insufficient samples" for a class.

A: This occurs when a class has fewer than 2 observations in either the original or synthetic group. The test is automatically bypassed and reported as "NA" without blocking other diagnostics.

Q: The KS test rejects several variables ($p < 0.05$). Should I abandon this method?

A: Not necessarily. For datasets with many variables (e.g., full-spectrum NIR with 256 wavelengths), some univariate rejections are expected by chance. If fewer than 10% of variables are rejected, resampling quality is generally considered adequate. If > 20% are rejected, consider parameter adjustments or a different algorithm.

9.4 Export and Downstream Use

Q: How should I use the exported balanced dataset in cross-validation?

A: Import the exported dataset into your modeling environment. Partition the data ensuring the test set contains only rows labeled "Original". The training set may include "Original" and "Synthetic" rows. Never expose synthetic samples to model evaluation metrics.

Q: The exported Excel file contains a "Type" column. Should I remove it before modeling?

A: Yes. The Type column (Original / Synthetic / Removed) is a metadata tag for traceability. Remove it before fitting any classification or regression model.

Q: My modeling software does not accept the "class" column notation from the exported file.

A: Rename the class column to the variable name expected by your modeling environment before importing. The Type and class columns can be manipulated freely after download.

10. Citation

If CWA: SS is used in published research, please cite the primary manuscript:

> Siqueira, J.C., Simões, P.H.C., Luna, A.S., Pinto, L. (2026). Chemometrics Web App: Synthetic Sampling. *Chemometrics and Intelligent Laboratory Systems*.

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